# Proposal: JUST Regulations Assistant

Team Project, Phase 1: Problem and Data · Building LLM Systems: From Models to Production · Jordan University of Science and Technology (JUST), Irbid

## 1. Team and roles

| Member | Phase 1 role | Responsible for |
|---|---|---|
| Hamza Moeen Rasheed Essa | Product lead + Evaluation lead | User and problem section; the 30 test questions, expected answers and scoring |
| Mohammad Ziad Abdelmajeed Al Rousan | Data lead | Collecting the 36 documents with links, checking extracted Arabic text and OCR numbers, data card |
| A'sem Zakariya Ali Elfrieh | Engineering lead | GitHub repository, text-extraction pipeline and tests, planned architecture on free compute |

Roles rotate at the start of every phase, so each member holds every role by the end of the project.

## 2. User and problem

**User:** JUST bachelor students, all majors. **Task:** finding the correct academic rule, for example absence limits, withdrawal deadlines, exam absence and excuses, grades and GPA, probation, postponement, graduation conditions and discipline.

**Today:** the rules are spread across separate official PDFs. Students open them one by one or ask friends, and a wrong answer can cost them grades, fees or a semester.

**Evidence:**

- **Fragmentation:** the official regulations page lists **346 separate PDF documents**, by title only, with no search across their contents. "Can I still withdraw?" needs the 17-page degree regulation.
- **Not searchable:** 17 of our 36 documents are scans or use broken fonts, so their text cannot be searched or copied; copying from the other 19 produces **895 corrupted words**.
- **Conflicting rules:** a medical excuse must reach the Health Center within **2 working days** (Medical Excuses Instructions, Art. 4), but the degree regulation allows **2 weeks** to submit it to the dean (Art. 7(e)).
- **Answers need several clauses:** 7 of the 27 answerable questions in our test set combine two or more clauses, and 4 of them combine clauses on different pages. For example, "which grade counts in my GPA after I retake a course?" needs Art. 9/4 (p. 10) together with the grade table in Art. 8(a) (p. 6).

## 3. What the system will do

A chat assistant for JUST bachelor students. Students ask in Arabic and the assistant answers in Arabic, **only from the official JUST documents**, citing the document and article with every answer. If the answer needs a personal record or a committee decision, or is not in the documents, it says «لا أعرف» ["I don't know"] and names the responsible office.

> **Student:** لحد أي أسبوع بقدر أنسحب من مادة نظرية؟ [Until which week can I withdraw from a theory course?]
> **Answer:** حتى 14 أسبوعاً من بداية الفصل (13 للمساقات العملية، وأول 7 أسابيع في الصيفي). [Up to 14 weeks from the start of the semester (13 for practical courses, the first 7 weeks in summer).]
> Source: Degree Regulation, Art. 12(b).

> **Student:** غبت عن الامتحان الفصلي، شو لازم أعمل؟ [I missed the midterm exam. What should I do?]
> **Answer:** تقدم عذراً لرئيس القسم الذي يطرح المساق خلال أسبوع؛ إذا قُبل يُعقد لك امتحان تكميلي، وإلا توضع لك علامة صفر. [Submit an excuse to the head of the department offering the course within a week; if accepted you get a makeup exam, otherwise a zero.]
> Source: Degree Regulation, Art. 8(b)/9.

> **Student:** هل رح تقبل اللجنة الطبية عذري؟ [Will the medical committee accept my excuse?]
> **Answer:** لا أعرف؛ قبول العذر قرار اللجنة الطبية الطلابية في المركز الصحي، ويمكن الاستعلام عن قرارها من خلال عميد كليتك. [I don't know; this is the Student Medical Committee's decision at the Health Center. You can ask about it through your dean.]

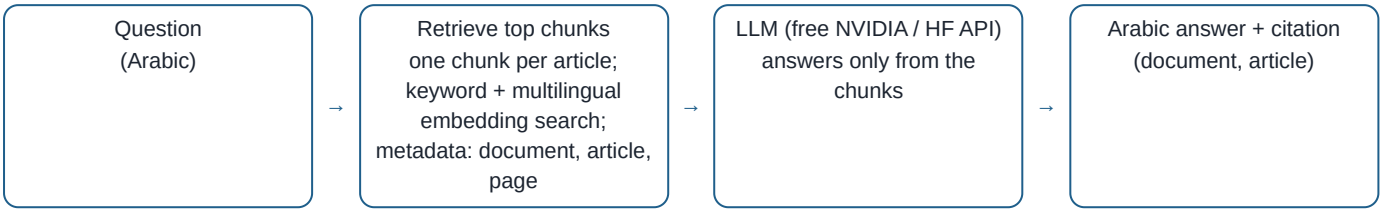
## 4. Data

| Source (official JUST regulations page) | Docs / pages | Status |
|---|---|---|
| Student rules: degree regulation, exams, medical excuses, discipline 2024, conduct code, fees, aid and loans, level exams, change of major, insurance, housing, weather suspension, and more | 24 / 73 | In hand, text extracted |

| | | |
|---|---|---|
| Rules mainly about university staff; a few matter to students (faculty office hours, student evaluation of faculty), the rest are kept but are not the focus | 12 / 34 | In hand, text extracted |
| **Planned next collection:** study plans and course catalogs for all bachelor majors (course codes, credit hours, prerequisites), published on www.just.edu.jo | not yet | Phase 2 |

In hand: 36 Arabic PDFs, 107 pages, about 80,000 tokens. Details, links and samples: data card and `data/SOURCES.md`.

## 5. Planned architecture




If retrieval confidence is low, or the question is personal or needs a decision, the system says «لا أعرف» ["I don't know"] and names the office. **Room to grow (RAG + agent):** later phases add tools the agent can call: an absence-limit calculator, a GPA-after-retake calculator, and a study-plan prerequisite checker. These are compared against plain RAG on the same test set.

## 6. How we will measure success

**Correct answer:** it states the same rule and every number (weeks, %, hours, GPA) as the expected answer, and nothing that contradicts the documents. **Numbers we report** (computed by `eval/score.py`), overall and by difficulty:

- **Answer accuracy**: manual review, with an LLM judge as a helper that a person checks.
- **Citation accuracy**: the citation names the right document and article.
- **Refusal accuracy**: out-of-scope questions answered «لا أعرف» ["I don't know"], and answerable questions not refused.
- **Retrieval recall@5**: the source article is among the top 5 retrieved chunks.

**Baseline:** keyword (BM25) retrieval + the same LLM and prompt. Every later version must beat it on the same 30 questions. Seed test set: 30 Arabic questions (15 easy, 10 medium, 5 hard; 3 refusals), answers written from the documents before running any model.

## 7. Risks

- **OCR errors in numbers and article numbers.** Backup: numbers in the test set come from text-layer pages or were checked against the PDF page image; key tables (grades, fees) are kept as hand-checked CSV.
- **Answers can depend on the major and the study-plan year.** Backup: every document and chunk carries metadata (major, plan year); when an answer depends on them, the system first asks the student's major.
- **Conflicting or outdated rules.** Backup: store the decision date with each chunk; when two rules differ, give both with their sources.
- **A wrong answer can cost grades or fees.** Backup: always cite the source; present the assistant as guidance, not an official decision.
- **Arabic quality and limits of free models; scope.** Backup: compare free models on the test set in Phase 2; add the study plans only after the regulations reach the baseline.